aucmat: matrix-first ROC analysis for high-dimensional biomarker screening in R

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Abstract

Summary: The area under the receiver operating characteristic curve (AUC) is the standard metric for evaluating continuous biomarkers against binary outcomes. When screening thousands of molecular biomarkers, researchers need matrix-wide AUC estimation with proper inference, multiplicity adjustment, and stability assessment. We present `aucmat`, an R package providing a matrix-first architecture: `aucmat(X, y)` computes direction-preserving AUCs with DeLong or stratified-bootstrap confidence intervals for every column of a numeric matrix in a single call. The package supports Benjamini-Hochberg multiplicity adjustment, three hypothesis-testing frameworks (superiority, non-inferiority, equivalence via TOST), an omnibus Wald test for equality of multiple correlated AUCs via the joint DeLong covariance matrix, and bootstrap rank-stability analysis. A two-stage Hurdle-AUC model handles zero-inflated data (scRNA-seq, microbiome). Eight simulation generators produce correlated biomarkers with controlled properties under exchangeable, AR(1), block, or user-supplied correlation structures with built-in calibration diagnostics. Over a dozen visualization functions are provided. The package is freely available under the MIT license at `https://github.com/vanhungtran/aucmat`.

Availability: `aucmat` v0.2.0, R $\geq$ 4.0.0, MIT license. `https://github.com/vanhungtran/aucmat` (source), `https://vanhungtran.github.io/aucmat` (documentation).
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1 Introduction

Receiver operating characteristic (ROC) analysis is the dominant framework for evaluating continuous biomarkers against binary outcomes in diagnostic medicine, prognostic modeling, and molecular screening [Hanley and McNeil, 1982, DeLong et al., 1988, Pepe, 2004]. The area under the ROC curve (AUC) summarises the probability that a randomly chosen positive subject has a higher biomarker value than a randomly chosen negative subject.

Modern molecular profiling technologies routinely measure thousands of features on a few hundred subjects. A proteomics panel may contain 5,000 proteins; a transcriptomics experiment 20,000 genes. The core task – identifying which biomarkers discriminate cases from controls – requires computing and testing an AUC for each feature, adjusting for multiplicity, comparing top candidates, and assessing ranking stability. Yet no single widely-used R package integrates these steps.

The R ecosystem provides excellent point solutions. `pROC` [Robin et al., 2011] is the gold standard: it computes DeLong confidence intervals using the Sun & Xu (2014) $\mathcal{O}(N \log N)$ algorithm [Sun and Xu, 2014], tests paired or unpaired AUC differences via `roc.test()`, provides partial AUC with bootstrap inference, binormal and density-smoothed ROC curves, and multiclass AUC via Hand & Till [Hand and Till, 2001]. However, pROC operates one biomarker at a time. `ROCR` [Sing et al., 2005] offers 28 performance measures and visualisation but no statistical inference. `precrec` [Saito and Rehmsmeier, 2017] adds fast C++-based precision-recall curves and confidence bands but provides no DeLong inference or multiplicity adjustment. `colAUC` [Tuszynski and Dietze, 2024] computes column-wise AUCs efficiently but provides no inference whatsoever. `dtComb` [Yerlitas et al., 2025] combines two biomarkers via 142 methods but does not perform matrix-wide screening. `cancerclass` [Budczies et al., 2024] builds classifiers from high-dimensional data but lacks DeLong-based screening.

`aucmat` fills this gap with a *matrix-first* design. The primary function `aucmat(X, y)` accepts $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a binary outcome $\mathbf{y} \in \{0, 1\}^n$, returning point estimates, standard errors, confidence intervals, raw and multiplicity-adjusted p-values, and diagnostic metadata for every biomarker in one call. Downstream functions for comparison, stability, cross-validation, panel construction, and power analysis consume the same S3 object, preserving outcome encoding throughout.

2 Methods

2.1 Direction-preserving AUC

For biomarker X and binary outcome $Y \in \{0, 1\}$, the AUC equals the Mann-Whitney U probability:

$$\mathrm{AUC} = P(X_i > X_j \mid Y_i = 1, Y_j = 0) \tag{1}$$

Most software reports $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$, silently reversing the biomarker. `aucmat` preserves the observed direction, reporting $\mathrm{AUC}_{\mathrm{raw}}$ (may be < 0.5), $\mathrm{AUC}_{\mathrm{strength}} = 0.5 + |\mathrm{AUC}_{\mathrm{raw}} - 0.5|$ (for ranking), and `effect_direction`. A tumour suppressor with $\mathrm{AUC}_{\mathrm{raw}} = 0.15$ discriminates as strongly as an oncogene with 0.85.

2.2 DeLong variance and joint covariance

The DeLong estimator [DeLong et al., 1988, Sun and Xu, 2014] uses placement values:

$$V_{10}(X_i) = \frac{1}{n_0} \sum_{j:Y_j=0} \left[\mathbf{1}(X_j < X_i) + \tfrac{1}{2}\mathbf{1}(X_j = X_i)\right], \quad i \in \mathcal{I}_1 \tag{2}$$

$$V_{01}(X_j) = \frac{1}{n_1} \sum_{i:Y_i=1} \left[\mathbf{1}(X_i > X_j) + \tfrac{1}{2}\mathbf{1}(X_i = X_j)\right], \quad j \in \mathcal{I}_0 \tag{3}$$

Let $S_{10} = (n_1 - 1)^{-1} \sum_{i \in \mathcal{I}_1} (V_{10}(X_i) - \bar{V}_{10})^2$ and $S_{01} = (n_0 - 1)^{-1} \sum_{j \in \mathcal{I}_0} (V_{01}(X_j) - \bar{V}_{01})^2$. The variance estimator is:

$$\widehat{\mathrm{Var}}(\widehat{\mathrm{AUC}}) = \frac{S_{10}}{n_1} + \frac{S_{01}}{n_0} \tag{4}$$

For p biomarkers on common subjects, the $p \times p$ joint covariance is:

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n_1}\widehat{\mathrm{Cov}}(\mathbf{V}_{10}) + \frac{1}{n_0}\widehat{\mathrm{Cov}}(\mathbf{V}_{01}) \tag{5}$$

This supports an omnibus Wald test of $H_0 : \mathrm{AUC}_1 = \cdots = \mathrm{AUC}_p$. With contrast matrix $\mathbf{C}$ (default: comparing biomarkers $2, \ldots, p$ against biomarker 1):

$$W = (\mathbf{C}\hat{\boldsymbol{\theta}})^\top (\mathbf{C}\widehat{\mathbf{\Sigma}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\boldsymbol{\theta}}) \sim \chi^2_r \tag{6}$$

using the effective rank r via eigenvalue tolerance. At $p = 2$, W equals the squared paired-DeLong z-statistic within numerical tolerance.

2.3 Multiplicity and stability

Screening p biomarkers produces p tests. `aucmat` applies Benjamini-Hochberg [Benjamini and Hochberg, 1995], Holm, or Bonferroni correction, reporting adjusted p-values as q-values. For TOST equivalence [Schuirmann, 1987], $p_{\mathrm{equiv}} = \max(p_{\mathrm{lower}}, p_{\mathrm{upper}})$ with adjustment on this combined value.

Bootstrap rank-stability (`auc_stability()`) resamples data B times (stratified by outcome), recomputes the full screening, and reports median/IQR rank distributions, top-k selection probabilities, and pairwise co-selection frequencies [Efron and Tibshirani, 1994].

2.4 Hurdle-AUC for zero-inflated data

Standard AUC assumes continuous biomarkers. On zero-inflated data (scRNA-seq, microbiome), 60%+ of values are zero, and standard AUC degrades to near 0.5. The Hurdle-AUC models zeros and expression as separate processes:

$$\text{Stage 1 (zero hurdle):} \quad P(X_j = 0 \mid Y) = \mathrm{logistic}(\beta_0 + \beta_1 Y) \tag{7}$$

$$\text{Stage 2 (magnitude):} \quad \mathrm{AUC}(X_j \mid X_j > 0) \text{ computed on expressed va} \tag{8}$$

The composite score for observation i is:

$$S_i = \begin{cases} \hat{p}_i & \text{if } X_{ij} = 0 \\ 0.5\,\hat{p}_i + 0.5\,\tilde{x}_{ij} & \text{if } X_{ij} > 0 \end{cases} \tag{9}$$

where $\hat{p}_i$ is the predicted non-zero probability from Stage 1 and $\tilde{x}_{ij}$ is the normalised biomarker value.

2.5 Multivariate simulation engine

The simulation engine is a core contribution of `aucmat`. We describe it in full mathematical detail, as it underpins the package's reproducibility guarantees and distinguishes it from ad-hoc simulation approaches.

2.5.1 The equal-variance binormal model

Under the standard binormal model [Hanley and McNeil, 1982], for biomarker $k \in \{1, \ldots, p\}$:

$$X_k \mid Y = 0 \sim \mathcal{N}(\mu_{k0}, \sigma^2), \qquad X_k \mid Y = 1 \sim \mathcal{N}(\mu_{k1}, \sigma^2) \tag{10}$$

The standardised mean separation $\delta_k = (\mu_{k1} - \mu_{k0})/\sigma$ determines the AUC via:

$$\mathrm{AUC}_k = \Phi\left(\frac{\delta_k}{\sqrt{2}}\right) \tag{11}$$

where $\Phi(\cdot)$ is the standard normal CDF. Inverting gives the closed-form calibration used by all `aucmat` simulators:

$$\delta_k = \sqrt{2} \cdot \Phi^{-1}(\mathrm{AUC}_k) \tag{12}$$

For AUC = 0.8, $\delta \approx 1.19$; for AUC = 0.9, $\delta \approx 1.81$. The class means are centred so the n-weighted grand mean is zero:

$$\mu_{k0} = \delta_k \cdot \left(-\frac{n_1}{n}\right), \qquad \mu_{k1} = \delta_k \cdot \left(\frac{n_0}{n}\right) \tag{13}$$

2.5.2 Class-conditional multivariate normal (`simulate_auc_matrix`)

For p biomarkers with correlation matrix $\mathbf{R}$, the class-conditional distribution generalises Eq. (10) to the multivariate case:

$$\mathbf{X} \mid Y = 0 \sim \mathcal{N}_p(\boldsymbol{\mu}_0, \mathbf{R}), \qquad \mathbf{X} \mid Y = 1 \sim \mathcal{N}_p(\boldsymbol{\mu}_1, \mathbf{R}) \tag{14}$$

where $\boldsymbol{\mu}_0 = (\mu_{10}, \ldots, \mu_{p0})^\top$ and $\boldsymbol{\mu}_1 = (\mu_{11}, \ldots, \mu_{p1})^\top$ follow Eq. (13). Both classes share the same $\mathbf{R}$.

The outcome Y is fixed *before* any biomarker is drawn. In `outcome_mode = "exact"`, $n_1 = \lfloor n\pi \rfloor$ observations are assigned $Y = 1$ with randomised order; in `"bernoulli"`, each $Y_i \sim \mathrm{Bernoulli}(\pi)$. When Y is user-supplied, its exact values and row order are preserved.

The matrix $\mathbf{R}$ is built from a named structure:

- **exchangeable**: $R_{ij} = \rho$ for all $i \neq j$. Requires $\rho > -1/(p-1)$ for positive definiteness.
- **AR(1)**: $R_{ij} = \rho^{|i-j|}$.
- **block**: $R_{ij} = \rho_w$ if i, j in same block, ρ_b otherwise. Block sizes must sum to p.
- **user**: R_{ij} directly from the supplied matrix.

When $\mathbf{R}$ is not positive definite, `feasibility = "nearest"` projects to the closest valid correlation matrix via eigenvalue clipping followed by renormalisation to unit diagonal. The adjustment is reported in `$feasibility` with Frobenius norm and maximum element-wise change.

After generation, each biomarker is standardised to zero mean and unit variance (an affine transform that preserves both AUC and Pearson correlation). Achieved correlation is computed on within-class centred residuals to avoid mixture-induced inflation (Simpson's paradox).

2.5.3 Gaussian copula with perturbation (`simulate_auc_copula`)

This simulator separates correlation control from discriminative signal:

Phase 1 – Gaussian Copula. For each class independently, n_c draws are taken from $\mathcal{N}_{p+1}(\mathbf{0}, \mathbf{R}_{\mathrm{work}})$, where $\mathbf{R}_{\mathrm{work}}$ is a $(p+1)\times(p+1)$ matrix combining the biomarker correlation $\mathbf{R}$ (upper-left $p \times p$) with latent outcome correlations $\boldsymbol{\rho}$ (last row/column). The latent correlations are approximated from target AUCs:

$$\rho_k^{\mathrm{latent}} = \frac{\delta_k}{\sqrt{1 + \delta_k^2}} \tag{15}$$

Each column is rank-transformed to uniform quantiles, then probit-transformed to standard normal scores. This copula step preserves rank correlations exactly regardless of marginal distributions.

Phase 2 – Iterative Perturbation. After Phase 1, each biomarker's empirical AUC, $\widehat{\mathrm{AUC}}_k$, is compared to its target. The gap $\Delta_k = \mathrm{AUC}_k - \widehat{\mathrm{AUC}}_k$ is translated to a mean shift via the binormal calibration:

$$\Delta\delta_k = \sqrt{2} \left[\Phi^{-1}(\mathrm{AUC}_k) - \Phi^{-1}(\widehat{\mathrm{AUC}}_k)\right] \tag{16}$$

The shift $\Delta\delta_k \cdot \sigma_k \cdot \eta^t$ is applied class-conditionally (positive class receives $+(n_0/n)$ times the shift, negative class $-(n_1/n)$), where $\eta \in (0, 1)$ is the step decay and t the iteration index. Perturbation stops when $|\Delta_k| < \tau$ for all k or after $T_{\max}$ iterations.

2.5.4 Hurdle simulator (`simulate_hurdle_auc`)

Generates zero-inflated biomarkers via a two-stage process. For biomarker j:

Stage 1 – Zero hurdle. Logistic parameters are solved from user-specified per-class zero rates:

$$\beta_0 = \mathrm{logit}(p_0), \quad p_0 = P(X_j = 0 \mid Y = 0) \tag{17}$$

$$\beta_1 = \mathrm{logit}(p_1) - \mathrm{logit}(p_0), \quad p_1 = P(X_j = 0 \mid Y = 1) \tag{18}$$

For each subject, $Z_{ij} \sim \mathrm{Bernoulli}(\mathrm{logistic}(\beta_0 + \beta_1 Y_i))$ determines whether $X_{ij} = 0$.

Stage 2 – Magnitude. For observations with $Z_{ij} = 0$ (expressed), values are drawn from a log-normal distribution with class-conditional means calibrated to achieve the target non-zero AUC via Eq. (12):

$$X_{ij} \mid X_{ij} > 0, Y_i = c \sim \mathrm{LogNormal}\big(\mu_c, \sigma^2\big) \tag{19}$$

where $\mu_1 - \mu_0$ is set by the binormal shift. Values are clipped to $[0, \infty)$.

2.5.5 Other simulators

`generate_data_probit()` uses a latent probit formulation where $\mathbf{X}$ and a latent $Y^* \sim \mathcal{N}(0, 1)$ are jointly $\mathcal{N}_{p+1}(\mathbf{0}, \boldsymbol{\Sigma})$, with $Y = \mathbf{1}\{Y^* > \Phi^{-1}(1-\pi)\}$. The full $(p+1) \times (p+1)$ correlation matrix simultaneously controls both biomarker-biomarker and biomarker-outcome associations. The latent correlations $\rho_k = \mathrm{Cor}(X_k, Y^*)$ are calibrated numerically via bisection over a large calibration sample ($n_{\mathrm{cal}} = 100{,}000$).

`generate_data_analytical()` constructs biomarkers sequentially from highest to lowest AUC. For $k \geq 2$, X_k is built as a linear combination of $X_1, \ldots, X_{k-1}$ (weighted by $\mathbf{R}_{k,1:k-1}\mathbf{R}^{-1}_{1:k-1,1:k-1}$ to achieve target correlations) plus an independent residual carrying the remaining mean separation. This is fast (closed-form, no calibration loop) but links AUC and correlation.

`generate_auc_vector()` constructs a single score with exact empirical AUC via rank assignment. Given n_1 positives and n_0 negatives, achievable AUCs form a discrete grid with step $\Delta_{\mathrm{AUC}} = 1/(n_1 n_0)$. The required rank sum is solved and normal quantile scores are assigned.

`generate_auc_cor_vector()` extends this with a strictly increasing Box-Cox transform tuned to achieve a target Pearson correlation without changing the AUC (since monotone transforms preserve ranks).

`simulate_auc_correlation()` repeatedly draws from the binormal model and records achieved (AUC, r) pairs to characterise their sampling distribution.

2.5.6 Simulation validation

`validate_simulation()` repeats a specification across R independent replicates with deterministic seed streams (seed $+\, r - 1$). For each replicate it records achieved AUCs and correlations, then reports:

$$\mathrm{Bias}_k = \frac{1}{R}\sum_{r=1}^{R}(\widehat{\mathrm{AUC}}_{kr} - \mathrm{AUC}_k) \tag{20}$$

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{R}\sum_{r=1}^{R}(\widehat{\mathrm{AUC}}_{kr} - \mathrm{AUC}_k)^2} \tag{21}$$

$$\mathrm{MCSE}_k = \frac{\mathrm{SD}_k}{\sqrt{R}} \tag{22}$$

plus target-interval hit rates for both AUCs and correlations. A bias large relative to MCSE signals a systematic calibration error.

3 Implementation

`aucmat` is implemented in pure R (6,200 lines, 34 source files). The primary function performs six steps: (1) input validation, (2) outcome normalisation (accepting logical, 0/1, factor, or character), (3) matrix-wide AUC via Mann-Whitney rank formulation, (4) DeLong or bootstrap inference, (5) multiplicity adjustment, (6) assembly with ranking and warnings.

All functions return S3 objects with `print()`, `summary()`, `plot()`, and `as.data.frame()` methods. Structured error classes allow programmatic handling. An RNG-safety mechanism (`with_seed()`) restores `.Random.seed` on exit.

Dependencies: pROC ($\geq$ 1.18), mvtnorm, ggplot2, rlang. Optional Suggests: GGally, glmnet, matrixStats, knitr, rmarkdown, ggrepel.

4 Usage

Basic screening:

```
fit <- aucmat(X, y, ci = "delong", adjust = "BH")
print(fit); summary(fit)
plot_auc_rank(fit, n_label = 4)
```

Paired comparisons with hypothesis testing:

```
compare_auc(fit, X, y, reference = "X1")  # superiority
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "noninferiority", margin = 0.05)
compare_auc(fit, X, y, reference = "X1",
  hypothesis = "equivalence", margin = 0.15)
compare_auc_global(fit, X, y)  # H0: all AUCs equal
```

Hurdle-AUC for zero-inflated data:

```
sim <- simulate_hurdle_auc(n = 400, prevalence = 0.3,
  target_hurdle_aucs = c(0.85, 0.72),
  zero_rate_neg = c(0.55, 0.80),
  zero_rate_pos = c(0.25, 0.70))
fit <- hurdle_auc(as.matrix(sim$data[,1:2]), sim$data$trut
# Standard AUC ~0.53 -> Hurdle AUC ~0.90
```

Cross-validation, panel construction, power analysis:

```
cv <- cv_aucmat(X, y, n_folds = 5, seed = 42)
panel <- fit_auc_panel(X[,top4], y, method = "ridge")
power_auc_matrix(target_aucs = 0.70, power = 0.80,
  prevalence = 0.3)
```

5 Benchmarking

Table 1: Empirical coverage of DeLong 95% confidence intervals, averaged across 27 simulation conditions (200 replicates each).

AUC	$n = 100$	$n = 300$	$n = 500$
0.65	0.898	0.937	0.957
0.75	0.915	0.943	0.933
0.85	0.883	0.923	0.905

Screening throughput. On a desktop workstation (AMD Ryzen 3.5 GHz, 32 GB RAM, R 4.5.1, Windows 11), `aucmat` screens 100, 1,000, and 10,000 biomarkers ($n = 500$, DeLong inference) in 0.54, 5.02, and 59.76 seconds. The equivalent per-column `pROC` loop requires 0.68, 6.42, and an estimated 64 seconds.

Simulation calibration. Across 27 conditions ($n = 100, 300, 500$; prevalence $= 0.2, 0.3, 0.5$; AUC $= 0.65, 0.75, 0.85$; 200 reps each), DeLong coverage ranged from 0.898 to 0.957, converging to 0.95 as n increased (Table 1). Power exceeded 0.85 for AUC ≥ 0.65 at $n = 100$ and reached 1.0 for AUC ≥ 0.75 at $n \geq 300$.

6 Comparison with existing packages

Table 2: Feature comparison of R packages for ROC analysis and biomarker screening. $+$ = supported; $-$ = not supported. Verified against CRAN documentation as of July 2026.

Feature	**aucmat**	pROC	precrec	ROCR	colAUC	dtComb
Matrix-first screening	+	−	−	−	+	−
Direction-preserving AUC	+	−	−	−	−	−
DeLong confidence intervals	+	+	−	−	−	−
Stratified bootstrap CI	+	+	−	−	−	+
Multiplicity (BH/Holm/Bonf)	+	−	−	−	−	−
One-sided alternative tests	+	+	−	−	−	−
Paired biomarker comparisons	+	+	−	−	−	+
Non-inferiority/equivalence	+	−	−	−	−	−
Omnibus Wald test (3+ AUCs)	+	−	−	−	−	−
Bootstrap rank stability	+	−	−	−	−	−
Partial AUC	+	+	+	−	−	−
Multivariate simulation	+	−	−	−	−	−
Simulation calibration	+	−	−	−	−	−
Cross-validated AUC	+	−	−	−	−	+
Penalised panel (ridge/lasso)	+	−	−	−	−	+
Hurdle-AUC (zero-inflated)	+	−	−	−	−	−
Precision-Recall curves	+	−	+	−	−	−
Power/sample size	+	+	−	−	−	−
Multiclass AUC (Hand-Till)	+	+	−	−	+	−
ROC smoothing	+	+	−	−	−	−
Group-stratified screening	+	−	−	−	−	−

Five capabilities are unique to `aucmat`: (i) matrix-first screening with direction-preserving AUC and multiplicity adjustment; (ii) non-inferiority and equivalence tests for paired AUCs; (iii) omnibus Wald test for three or more correlated AUCs; (iv) bootstrap rank-stability analysis; and (v) Hurdle-AUC for zero-inflated data.

For a single biomarker with smoothed ROC curves, `pROC` remains the standard. For speed on $> 10^5$ columns, `colAUC` is recommended. For precision-recall of imbalanced data, `precrec` is preferred. For combining two biomarkers, `dtComb` provides the most exhaustive methods. For end-to-end biomarker screening with full inference, `aucmat` is the recommended tool.

7 Discussion

`aucmat` provides a unified, statistically principled framework for biomarker screening. Its matrix-first design reduces the code burden from dozens of scripting steps to a few function calls.

Limitations. The current version (0.2.0) is restricted to binary outcomes for the primary workflow (multiclass is experimental). DeLong requires common subjects; unpaired comparisons are not implemented (pROC provides this). Partial AUC is an internal engine not yet exposed in `aucmat()`. The DeLong estimator is slightly liberal at small n (coverage ≈ 0.90 at $n = 100$), consistent with its finite-sample behaviour; bootstrap intervals are recommended when $n < 100$ per class.

Future work. v0.3: nested cross-validation for penalised panel models. v0.4: multiclass AUC with one-versus-rest, one-versus-one, and Hand-Till estimators. v0.5: cumulative/dynamic time-dependent AUCs for survival outcomes [Heagerty et al., 2000, Blanche et al., 2013]. v0.6: sparse-matrix and on-disk backends for genomics-scale data.

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